A Weave of Two Records

A storyteller's question, followed into mathematics

Dritëro M. · October 2026

Part I is an overview. Part II is a technical appendix. Every calculation is public at `https://github.com/originaxiom/origin-axiom`.

Part I An overview

Where it started

I am a filmmaker. My training is in drama, where a story exists only when someone listens to it, and changes with who listens. Watching physics explained well, I began to ask a storyteller's question about reality: what is the least a story needs in order to exist at all?

My working answer became a principle: *existence is what remains when cancelling to nothing cannot complete.* Something is always left over, and what is left over is a description that never runs out. This document is about what happens when that sentence is taken literally, as mathematics, and followed as far as it goes.

The idea in one picture

The simplest precise version I found has two counters, call them a and b, and two ways to update them: add a to b, or add b to a. Any description is a sequence of these two moves, and the two never cancel each other out.

One rule stands out among all sequences: replace a by ab and b by a. Repeated, it writes a word that never repeats, $abaababaabaab\ldots$, and it grows by the golden ratio, $\varphi \approx 1.618$. I call one application of the rule a *tick*. Two ticks, read as geometry, build a well-known three-dimensional space: the complement of the figure-eight knot, one of the two simplest spaces of its kind.

One object, or all of them

For a long time I studied that single space. Then the project changed course. Instead of picking one object, it follows every object the two moves allow, all at once. We call that the *weave*, and we only trust a result if it holds for all of them. That change mattered more than any single calculation, because many earlier "results" turned out to be true of one object and not of the rest.

How the work is done

I am not a trained physicist, so the method had to protect me from myself.

- The work runs as several AI-assisted research lanes that check one another. A result from one lane counts only after another has recomputed it its own way.
- Before any computation runs, its predictions are written down and sealed. Afterwards the result is graded against them, and failures are kept as carefully as successes.
- Everything is public: the code, the sealed predictions, the failures and the corrections. That is about 1,400 recorded computations and more than 7,000 automated checks so far.
- I make choices only where the mathematics cannot, and every such choice is written down as an assumption.

What came out, in plain words

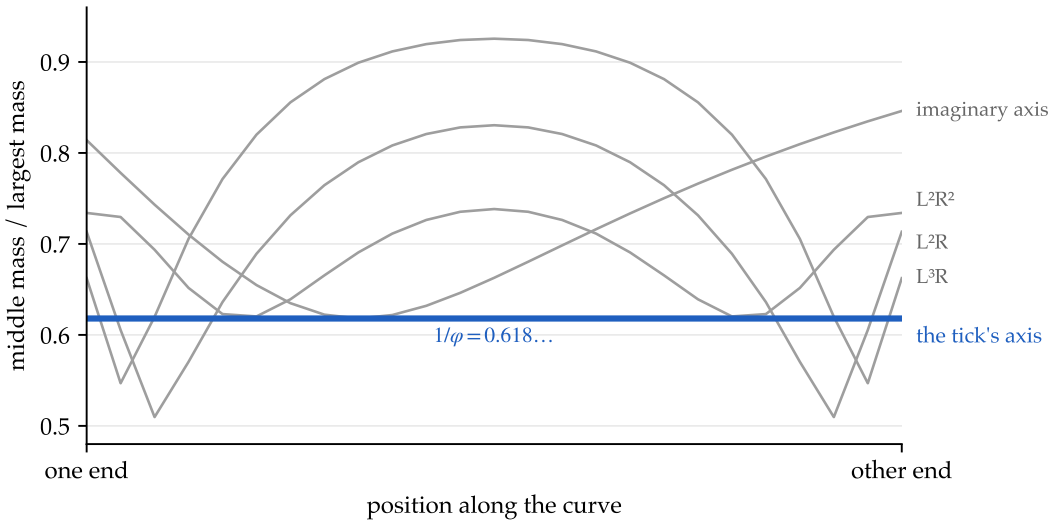
1. A forced three. Two counters can be "odd" in exactly three ways: a odd, b odd, or both. The moves shuffle these three, and no single move owns them. Out of this the mathematics produces three identical slots with a built-in handedness, a preference for one mirror image (which hand is called "left" is a convention, since a single tick swaps them). Physics has a pattern like this: three generations of matter, alike except in mass, and left–right asymmetric. The three is derived. Calling the slots *the* generations needs one assumption, which I state openly. The forces of the Standard Model are not derived: they appear only if one chooses them.

2. The slots are old friends of mathematics. The way the three slots transform under the moves turned out to be exactly the way three classical functions transform: Jacobi's theta functions, studied since the 1820s. The match is exact, with no adjustment. Physicists who build models of particle families from these symmetries already use the same three functions. What surprised me is that the principle's own moves land on them by themselves.

3. Where masses would come from, and a golden surprise. Particle masses would come from one remaining "shape" number of the geometry, called τ. Nothing in the principle has fixed it yet, so no mass is predicted. But the tick moves that shape number along one particular path, its own axis. Along that path the simplest mass pattern the structure allows is

$$1 \ : \ 0.618\ldots \ : \ 0.382\ldots \ = \ 1 : \varphi^{-1} : \varphi^{-2},$$

the golden ratio again. It is exact at one point of the path, and holds to fifteen decimal places at every other point we computed; a proof for the whole path is still open. It holds on no other path we checked. We found it by surprise, and only then tested it against controls written down in advance. The figure shows it.



Along the tick's own axis (blue) the ratio never moves; along other paths (grey) it does.

This is not the pattern of real particle masses, which are far more spread out. What it shows is the first place where the principle, by itself, fixes a ratio.

What it is not

This is not a theory of everything. The Standard Model needs 19 numbers as input: the masses of the quarks and charged leptons, the angles and phase that mix the quarks, the strengths of the three forces, and a few more. This work fixes none of them. Several results rest on stated assumptions, listed in the appendix. Many of our results are negative, and I think those matter just as much, because they show where the structure has nothing to say.

Why I am sending it

Your videos made physics feel like a story worth following. I would be grateful for an honest reaction. Does the split between *structure*, which the principle seems to give, and *numbers*, which it does not, sound meaningful to you? And where would a physicist push first? The appendix gives the technical version, and every calculation is in the public repository: `https://github.com/originaxiom/origin-axiom`.

Part II Technical appendix

Summary. A two-letter rewriting system, read as mapping classes of a once-punctured torus, is followed over every object it allows at once. Its moves single out a three-dimensional representation T, whose multiplier is exactly that of $\eta^{21}(\theta_2^2, \theta_3^2, \theta_4^2)$. This is the level-4, metaplectic structure already used in modular flavour models; what is reported here is that the system's own moves land on it with no choice of representation. The modulus τ of the torus is then the only thing that breaks the symmetry, so mass couplings are vector-valued modular forms, and their number at each weight is counted. One result appears to be new. On the axis of the system's rule, the lowest coupling has normalized masses $1 : \varphi^{-1} : \varphi^{-2}$. This is proved at one point and holds to machine precision at every point computed; a proof along the whole axis is open. None of the Standard Model's 19 parameters is fixed. Each claim carries its grade. *Derived* means computed here. *Registered* means computed by an independent lane and not yet recomputed here. *Conjectured* means numerical only. *Given* names an assumption. Arc numbers (B1600–B1636) point to the public record.

1 The principle and its grammar

Postulates. Existence is what remains when cancelling to nothing cannot complete. The remainder is an inexhaustible description. The description is read as mathematical structure.

Grammar (postulated). Two integer records are updated by the unit shears $L = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ and $R = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$, acting on $F_2 = \langle a, b \rangle$ by $L : (a, b) \mapsto (a, ab)$ and $R : (a, b) \mapsto (ab, b)$, as mapping classes of the once-punctured torus. Reading "cancelling cannot complete" as the non-vanishing commutator of a and b is an interpretation, not a derivation.

The rule. The Fibonacci substitution $\sigma : a \mapsto ab,\ b \mapsto a$ is one *tick*. Its matrix is $LP = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, with P the swap of the two records, and two ticks give $\sigma^2 = LR = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ (trace 3, eigenvalues $\varphi^{\pm 2}$). The mapping torus of LR is the figure-eight knot complement. A uniqueness theorem in the record shows that seven axioms of a minimal record-transfer system single out LR, machine-checked from 144 candidates to 1. The record itself grades this as a consistency demonstration, not a search: the object was found before the axioms were written, so the result is a conditional reconstruction of a chosen sector, not a derivation from nothing.

Rulings. The swap P is part of the act but not a symmetry of what is observed (only even numbers of ticks are observed). Counts are non-negative, which excludes $-I$. The observed symmetry is therefore the group generated by L and R. Using that group, with inverses, assumes that inverse moves are legal, which the record lists as an open fork. We call this setting the *ruled branch*.

2 The weave

A *thread* is the mapping torus of one positive word in L and R. The *weave* is the joint action of all the moves on the two records. A result is accepted only if it holds on every thread under a stated rule, none chosen.

The common point. The quaternion representation $\rho_Q : a \mapsto i,\ b \mapsto j$ is the only point of the character variety fixed by every move. On each odd-trace thread it is that thread's own geometry reduced modulo a prime of norm 3 (B1601).

Three parity sectors. The three non-trivial characters χ_p of $H_1(F_2; \mathbb{Z}/2)$ are permuted by the

moves: each move fixes one and swaps the other two. Over ρ_Q,

$$V = \bigoplus_p H^1(F_2; \chi_p \otimes \rho_Q) = T \oplus \bar{T}, \qquad \dim T = 3,$$

with one line of T for each χ_p. The lifts of L and R generate a group of order 96, irreducible on T. The *clock's line* belongs to the character that is -1 on both letters, which reads the parity of a word's length.

3 Three: what is derived

Two senses of "chiral" are kept apart below. *Weave-chiral* means distinguished from its mirror under the weave's own group. *Gauge-chiral* means carried by a complex representation of a gauge group, which is what the Standard Model needs.

Item	Result	Grade
A flavour triplet	T is three-dimensional and irreducible under the weave's group: three alike sectors	derived
As the generations	the three sectors are the three generations of matter	a reading, given Λ (§7)
Weave-chirality	a Hodge–Riemann form kept by L and R is positive on T and negative on $\bar{T}$; the three holomorphic zero modes span T; the rule σ itself exchanges T and $\bar{T}$, so which hand is "left" is a convention	derived, given the even-ticks ruling
At the puncture	every boundary condition the moves keep has index ± 1 or ± 3, never 0; with the parity grading only ± 3	derived
Four-dimensional count	0 at weight 3/2 without data at the cusp (§4); ± 3 with one unit there	0 derived; ± 3 registered, given Λ and the unit
CP	the generalised CP is the swap: CP violation allowed, not forced	derived
Gauge group and gauge-chirality	$S(U(3) \times U(2))$ appears only by choosing a frame through E_6	a selection, not derived

The 19 parameters are the six quark masses, the three charged-lepton masses, the four parameters of the quark mixing matrix (the 13 *flavour* parameters), the three gauge couplings, the strong CP angle and the two Higgs parameters. None is fixed. Neutrino masses and mixing lie outside the 19.

4 The three sectors and the three even theta constants

The moves act on the modulus τ: L as $\tau \mapsto \tau + 1$, and $RL^{-1}R$ as $\tau \mapsto -1/\tau$. The lifts on T obey the braid relation exactly, with $\Delta^4 \mapsto -1$ for the half twist $\Delta = LR^{-1}L$. So they form a genuine representation of $\mathrm{Mp}_2(\mathbb{Z}) = B_3/\langle \Delta^8 \rangle$. With S and T fixed by the action on homology, this representation equals the multiplier of

$$F(\tau) = \eta(\tau)^{21}(\theta_2(\tau)^2,\ \theta_3(\tau)^2,\ \theta_4(\tau)^2)$$

with no rescaling. A monomial intertwiner sends θ_3^2 to the clock's line, and θ_2^2, θ_4^2 to the other two. The T-exponents are $\frac{1}{8}, \frac{3}{8}, \frac{7}{8}$ (B1634; derived).

The space spanned by $\theta_2^2, \theta_3^2, \theta_4^2$ is the space of weight-1 forms of level 4. It is the known weight-1 triplet of the double cover of modular S_4, set in the metaplectic, half-integral-weight framework of modular flavour models. The new part is the route: the system's own moves land on this representation. (This record first read the lifts as only projective. The cause was a sign fixed inside a helper routine, found by auditing that negative.)

Consequence. Take weight 3/2, the weight the link Λ's Dirac operator fixes for the zero modes (an independent lane's computation, registered). The Riemann–Roch number of T is $\chi_{3/2} = 0$. This comes from the Borcherds–Skoruppa formula, and independently from counting forms in the ring of $\Gamma(4)$, the polynomials in $\theta_2^2, \theta_3^2, \theta_4^2$ modulo $\theta_3^4 = \theta_2^4 + \theta_4^4$. The two agree at every weight from $-\frac{1}{2}$ to $\frac{27}{2}$ (derived). So a four-dimensional count of three needs one unit of end data at the cusp, a tagged postulate (§7).

5 What breaks the symmetry, and the count

By Schur's lemma, an invariant mass term on the irreducible T is degenerate. On the ruled branch, the moves fixing a point τ act on T as follows (B1634; derived):

τ	stabilizer on T	effect on couplings
generic	four scalars (powers of Δ^2)	none
i	cyclic of order 8	depends on the weight mod 4
$\omega = e^{2\pi i/3}$	cyclic of order 12	every coupling degenerate

The inner automorphisms by a and b are not words in L and R, because $\langle L, R\rangle \cong B_3$ meets $\mathrm{Inn}(F_2)$ only in $\langle\Delta^4\rangle$ (checked to word length 12). An earlier theorem, that mixing in τ alone is a permutation, assumed them as symmetries. It holds only where they act, with the swap or the sign as moves. So on the ruled branch τ **is the state that breaks the weave's group**.

Couplings. With a flavour-blind Higgs, a mass term is a matrix $Y(\tau)$ of modular forms of weight k, equivariant for $T \otimes T$. At each k these form a finite-dimensional space (B1635; two independent routes):

k	1	3	5	7	9	11
symmetric part (diagonal + off-diagonal)	1 (0+1)	1 (0+1)	3 (1+2)	3 (1+2)	5 (2+3)	5 (2+3)
all of $T \otimes T$ (adds antisymmetric)	1	2	4	5	7	8

At generic τ, symmetric couplings give three distinct masses, and two sectors mix by a non-permutation. An antisymmetric coupling always has singular values $(s, s, 0)$. Of that arc's six sealed predictions, two failed: a non-degeneracy threshold, and "every coupling is degenerate at i".

The count. Given each sector's weight k_s, the free real parameters are $2 + \sum_s (2d(k_s) - 1)$: the real and imaginary parts of τ, plus each sector's coefficients up to a phase. Kähler (kinetic) terms are taken minimal. At $k = 1$ and $k = 3$ the symmetric couplings are purely off-diagonal, and the sum rule of §6 rules that configuration out for charged fermions. An independent lane takes zero modes of weight 3/2 and a weight-0 Higgs, so $k = 3$ for general $T \otimes T$. It counts 11 of the 13 flavour parameters free in the Standard Model frame, with two predicted

relations among quark masses and mixing (registered; a sealed fit to data is pending). Nothing canonical fixes τ. $\prod_p |\theta_p/\eta|^2 = 4$ is constant (Jacobi). The independent lane finds energy-like functionals minimal at ω, where couplings are degenerate, and partition-like ones running to the cusp (registered).

6 Golden masses on the axis of the rule

The lowest coupling. At $k = 1$ the symmetric coupling is unique up to scale (B1636; derived numerically to 3×10^{-13}):

$$Y(\tau) \;\propto\; \begin{pmatrix} 0 & \theta_4^2 & \theta_3^2 \\ \theta_4^2 & 0 & \theta_2^2 \\ \theta_3^2 & \theta_2^2 & 0 \end{pmatrix}.$$

Sum rule (a theorem). A complex symmetric 3×3 matrix with zero diagonal can be rephased to positive entries. Its eigenvalues then sum to zero with signs $(+, -, -)$, so its singular values obey $m_1 = m_2 + m_3$.

One number decides the spectrum. Since $\theta_2\theta_3\theta_4 = 2\eta^3$, the normalized spectrum depends on τ only through

$$F(\tau) = \sum_{p=2,3,4} \Big|\frac{\theta_p(\tau)}{\eta(\tau)}\Big|^4 \;\geq\; 3 \cdot 4^{2/3}.$$

The minimum is at ω, with masses $(1, \frac{1}{2}, \frac{1}{2})$. $F = 8$ gives the characteristic polynomial $\lambda^3 - 2\lambda - 1 = (\lambda + 1)(\lambda^2 - \lambda - 1)$, that is, masses $1 : \varphi^{-1} : \varphi^{-2}$.

The axis. The tick σ is a glide reflection, and $\sigma^2 = LR$ a translation, along the same geodesic: the semicircle $|\tau - \frac{1}{2}| = \frac{\sqrt{5}}{2}$ through i. It is the lowest closed geodesic of the weave (B1629). At $\tau = i$ the golden spectrum is exact, because $\theta_2(i) = \theta_4(i)$ and $\theta_3(i)^2 = \sqrt{2}\,\theta_2(i)^2$. Along the whole axis it holds at every point computed (66 points, deviation 3×10^{-15}). The result was found unpredicted, then tested with controls sealed beforehand. Five other closed geodesics, three reflection lines and six other couplings all vary.

Conjecture. The level set $F = 8$ is exactly the union of the geodesics of discriminant 5, the $SL_2(\mathbb{Z})$-images of the axis: 84 of 88 sampled points lie on them to 10^{-9}, and the other 4 sit at crossings. Discriminant 5 is that of $x^2 - x - 1$, the golden polynomial. A short literature search found mass ratios built from level-4 theta constants at a modulus on the imaginary axis (Barger, 2026), golden-ratio *mixing* from A_5 symmetries, and golden *mass relations* between sectors. It found no golden spectrum on a geodesic.

Limits. Measured quark and charged-lepton masses violate the sum rule by factors of 17 to 136, so this coupling alone is no charged sector's mass matrix. Whether the principle places τ on the axis is open. A hand calculation gives golden eigenvectors at i (one maximal angle, and one with $\tan\theta = 1/\varphi$), not yet verified by computation.

7 Assumptions

Assumption	Content	Status
The principle	the three postulates of §1	postulated
The grammar	two integer records, the shears L, R, reversible updates, aperiodicity, the once-punctured torus	postulated
Inverse moves	used to form the group L and R generate	open fork
Even ticks observed	the swap is not a symmetry of what is observed	owner's ruling
Positivity	non-negative counts; $-I$ excluded	owner's ruling
The link Λ	a generation is a field on spacetime × the torus in the 27 of E_6 carrying the weave's bundle, one per parity; holomorphic zero modes are left-handed matter. The 27 brings exotic states, and anomalies and proton decay are among the triggers to revisit it	tagged postulate
The cusp unit	one unit of end data at the cusp	tagged postulate
τ and the weights	not fixed ($\tau = \omega$ tested and retired); whether τ is a physical modulus is open	open

8 Open questions and tests

Still open:

- what places τ;
- the sectors' weights;
- a forced complex structure on the gauge side;
- a proof of the golden identity;
- any value.

Test	Prediction	Status
Lepton mixing: the TM1 reading ($\bar{T} \otimes T$, the weave's full group as flavour, residual subgroups RL and RRL selected; a vacuum-alignment reading, not the τ-breaking of §5)	$\sin^2\theta_{12} = 1 - \frac{2}{3\cos^2\theta_{13}} \approx 0.318$; $\delta_{CP} \in [252.7°, 292.9°]$ or its mirror $[67.1°, 107.3°]$ given θ_{23}	0.85σ (NuFIT 6.0); the TM2 reading ($T \otimes T$) at 2.74σ, and at 3.64σ with JUNO's first result
Sum rule	zero-diagonal symmetric sectors obey $m_1 = m_2 + m_3$	excludes them for charged fermions
Two quark relations at $k = 3$	8 quark shape numbers from 6 parameters (registered count)	sealed fit pending

9 Method

For each computation, the cells, predictions and priors are committed and hashed before it runs. Results are graded against the seal, and failed cells are reported as failed. Negatives are audited for bugs, inputs and integrity before anything is built on them. Several AI-assisted lanes, under one owner, check one another through separate code paths. A claim from one lane is registered, not banked, until another recomputes it. The foundation document changes only through recorded amendments (now v1.41). The record is public: about 1,400 computations and 7,000 automated checks. The figure in Part I was recomputed for this document from the theta series; the sealed run is B1636.

References

- The record: `https://github.com/originaxiom/origin-axiom` (GENESIS.md v1.41; docs/THE_WEAVE.md; arcs B1600–B1636 under frontier/).
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